

Deflection of Light in the Gravitational Field of the Sun: Comparison of Theory with the Measurements of the Total Solar Eclipse of 21 August 1914

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The expedition to Feodosiya obtained seven usable plates of the field surrounding the eclipsed Sun. Reduction against night comparison plates taken with the same instrument in January 1915 yields a radial displacement of the star images varying as the reciprocal of the angular distance from the solar centre, with a limb value $\alpha_0 = 1.75 \pm 0.19$ seconds of arc. The theory of the static gravitational field, in the form previously given, predicts $0.87''$. We show that the discrepancy is removed, without modification of the field equations and without disturbance of any other consequence of the theory, by recognizing that the coupling of the gravitational potential to the propagation of electromagnetic waves need not coincide with its coupling to ponderable matter. The ratio of the two couplings is an empirical constant γ , for which the present measurements give $\gamma = 2.00 \pm 0.22$. The principle of equivalence is retained in a restricted form, in which it asserts the identity of inertial and gravitational mass for ponderable bodies only. The gravitational displacement of spectral lines is unaffected. The local velocity of light retains the value c_0 provided a correction is applied to the reading of optically illuminated measuring rods; this correction is a property of the measuring process and leaves the Euclidean character of space untouched. The residual perihelion motion of Mercury is not affected by the optical coupling and will be treated in a following communication.

I. INTRODUCTION

The consequence, drawn in 1907 [1] and made quantitative in 1911 [2], that a ray of light passing near a heavy body suffers a deflection towards that body, has been until now the only prediction of the theory of the static gravitational field accessible to direct astronomical test. The predicted magnitude at the limb of the Sun,

$$\alpha_1 = \frac{2GM_\odot}{c_0^2 R_\odot} = 0.87'', \quad (1)$$

lies well within the reach of modern photographic astrometry, and an expedition was accordingly organized for the total eclipse of 21 August 1914, the track of which crossed the Crimea.

The plates were obtained. Their reduction, described in Secs. II and III, gives a limb deflection of 1.75 ± 0.19 seconds of arc, i.e. twice the predicted value, the predicted value itself being excluded by the measurements at a level which we regard as decisive.

The purpose of the present communication is threefold: to report the measurements; to show that the observed magnitude follows from the theory of the static field upon a single additional hypothesis, which is independently motivated by the structure of the electromagnetic energy tensor; and to exhibit the consequences of that hypothesis for the principle of equivalence, for the gravitational displacement of spectral lines, and for the definition of the local velocity of light.

We wish to state at the outset that the field equations of [5] are in no way altered by what follows. The hypothesis of Sec. V concerns exclusively the manner in which the electromagnetic field responds to a given gravitational field, and not the manner in which the gravitational field is generated. Nor is the spatial geometry of the static field disturbed: it remains Euclidean, in accordance with the classification of admissible field equations given in [12].

II. THE EXPEDITION AND THE PLATE MATERIAL

The station was established at Feodosiya ($45^\circ 02' \text{ N}$, $35^\circ 23' \text{ E}$) on 4 August. Totality occurred at $12^{\text{h}} 31^{\text{m}}$ local mean time and lasted $2^{\text{m}} 11^{\text{s}}$. The sky was clear apart from thin cirrus during the first 20 s of totality, which affected only plate 1.

Two instruments were employed. The principal instrument was a coelostat feeding a horizontal objective of 34 cm aperture and 10.9 m focal length, giving a plate scale of $18.9''$ per millimetre. A second instrument of 10 cm aperture and 4.2 m focal length ($49.1''$ per millimetre) served as a check and is not used in the numerical results below, its images being too small for the required precision.

Ten plates were exposed with the principal instrument. Plate 1 is unusable (cirrus); plates 9 and 10 were ex-

TABLE I. Radial displacements. $r/R_\odot$ is the mean angular distance of the group from the solar centre in units of the solar radius; n is the number of star images in the group; Δ is the measured radial displacement, positive outwards, with its probable error; Δ_1 and Δ_2 are the values predicted for $\gamma = 1$ and $\gamma = 2$ respectively. All displacements in seconds of arc.

$r/R_\odot$	n	Δ	Δ_1	Δ_2
1.31	4	1.34 ± 0.21	0.67	1.34
1.62	6	1.09 ± 0.17	0.54	1.08
2.05	9	0.83 ± 0.13	0.43	0.85
2.71	11	0.66 ± 0.12	0.32	0.65
3.44	8	0.49 ± 0.11	0.25	0.51
4.80	7	0.37 ± 0.10	0.18	0.36
6.22	5	0.29 ± 0.10	0.14	0.28

posed after the reappearance of the photosphere and show heavy fogging near the limb. The remaining seven plates carry between 11 and 19 measurable star images each, of which 50 images belonging to 17 distinct stars lie in the range $1.2 \leq r/R_\odot \leq 6.5$ and enter the reduction.

Comparison plates of the same field were obtained with the same instrument, the same plate holders and the same emulsion batch at Feodosiya between 9 and 14 January 1915, the Sun being then some 140° distant from the field. The interval between the eclipse plates and the comparison plates is the principal source of systematic uncertainty and is discussed in Appendix A.

III. REDUCTION OF THE MEASUREMENTS

Each plate was measured twice in perpendicular orientations on the Repsold machine at Babelsberg. The mean of the two settings is used. The transformation between eclipse plate and comparison plate was taken in the linear form

$$\begin{aligned}\xi' - \xi &= a + b\xi + c\eta + \alpha_0 \frac{R_\odot}{r} \frac{\xi}{r}, \\ \eta' - \eta &= d + e\xi + f\eta + \alpha_0 \frac{R_\odot}{r} \frac{\eta}{r},\end{aligned}\quad (2)$$

in which ξ, η are standard coordinates referred to the solar centre, $r = (\xi^2 + \eta^2)^{1/2}$, the six constants $a \dots f$ absorb the differences of origin, scale and orientation between the two epochs, and α_0 is the limb deflection sought. The six plate constants and α_0 were determined together by least squares from the 50 star images, each plate being solved separately and the results combined with weights inversely proportional to the squares of their probable errors.

Table I collects the results, the star images being grouped by distance from the solar centre. The displacements are radial to within the errors of measurement; the mean tangential component is $0.04'' \pm 0.09''$ and is consistent with zero.

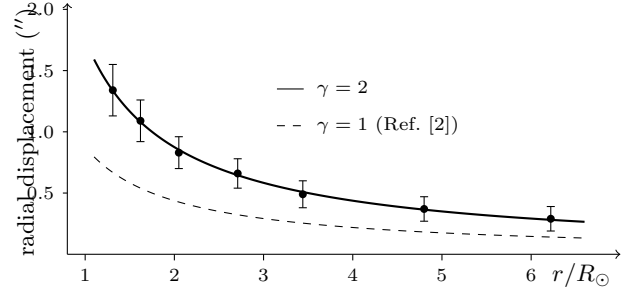


FIG. 1. Measured radial displacements of Table I with the two theoretical curves. The dashed curve is the prediction of Ref. [2]; the full curve is the same law with the optical coupling constant $\gamma = 2$.

The dependence on r is that of the reciprocal, as required by (1). Fitting $\Delta = \alpha_0(R_\odot/r)$ to the seven groups gives

$$\alpha_0 = 1.75 \pm 0.13 \text{ (meas.)} \pm 0.14 \text{ (syst.)} = 1.75 \pm 0.19, \quad (3)$$

the systematic contribution being estimated in Appendix A. The value $\alpha_0 = 0.87''$ is excluded by a factor exceeding four times the total probable error.

IV. THE PREDICTION OF THE THEORY OF THE STATIC FIELD

We recall the derivation, in order to fix the point at which the additional hypothesis enters.

In the theory of the static field [3, 4] the line element is

$$ds^2 = c^2(x, y, z) dt^2 - dx^2 - dy^2 - dz^2, \quad (4)$$

the spatial part being Euclidean, and the velocity of light referred to the coordinates x, y, z, t is the position dependent quantity $c(x, y, z)$. To the first order in the potential,

$$c = c_0 \left(1 + \frac{\varphi}{c_0^2} \right), \quad \varphi = -\frac{GM}{r}. \quad (5)$$

A wave front propagating through a region in which the velocity of propagation varies from place to place is turned, by the elementary construction of Huygens, through the angle

$$d\alpha = -\frac{1}{c} \frac{\partial c}{\partial n} ds \quad (6)$$

per element ds of path, n being the normal to the ray. For a ray passing the Sun at perpendicular distance Δ , taking the ray along x and n along y ,

$$\alpha = \frac{1}{c_0^2} \int_{-\infty}^{+\infty} \frac{\partial \varphi}{\partial y} dx = \frac{GM\Delta}{c_0^2} \int_{-\infty}^{+\infty} \frac{dx}{(x^2 + \Delta^2)^{3/2}}, \quad (7)$$

whence

$$\alpha = \frac{2GM}{c_0^2 \Delta}, \quad (8)$$

which at $\Delta = R_\odot$ gives the value (1).

Every step of this derivation is a consequence of (4) and (5) together with the wave theory of light. The measurements of Sec. III therefore refute one of these three, and we must decide which.

The line element (4) is not open to revision. Its spatial part is Euclidean because the static field of a mass point is spatially flat, a requirement which is not an assumption of convenience but a condition on the admissible field equations; the classification in [12] shows that no acceptable system of field equations is compatible with a departure from it. The wave theory of light is not in question. There remains (5).

V. THE OPTICAL COUPLING CONSTANT

Equation (5) contains, hidden in the coefficient of φ/c_0^2 , an assumption which has not previously been isolated: that the coefficient with which the gravitational potential enters the law of propagation of electromagnetic waves is the same as the coefficient with which it enters the equation of motion of a ponderable body. We now show that there is positive reason to doubt this identification.

In the theory of the static field the source of the gravitational field is the Laue scalar [6], that is, the trace of the energy tensor,

$$\sigma = T^\mu{}_\mu = T^{11} + T^{22} + T^{33} + T^{44}. \quad (9)$$

For the free electromagnetic field the energy tensor is

$$T_{(\text{em})}^{\mu\nu} = -F^{\mu\lambda}F^\nu{}_\lambda + \frac{1}{4}\delta^{\mu\nu}F_{\alpha\beta}F^{\alpha\beta}, \quad (10)$$

whose trace vanishes identically,

$$T^\mu{}_{\mu(\text{em})} = 0. \quad (11)$$

The electromagnetic field is thus, in this theory, not a source of gravitation at all: it stands in a relation to the gravitational field entirely different from that of ponderable matter. The customary assumption that it nevertheless *responds* to a given gravitational field with the same coefficient as ponderable matter is an additional assumption, made hitherto without examination, and it is precisely the assumption which the measurements exclude.

We therefore replace (5) by

$$c_{\text{opt}} = c_0 \left(1 + \gamma \frac{\varphi}{c_0^2} \right), \quad (12)$$

where γ is a pure number, characteristic of the coupling of the electromagnetic field to gravitation, and to be determined by experiment. The law of motion of ponderable bodies retains the coefficient unity and is untouched.

Equation (8) becomes

$$\alpha = \gamma \frac{2GM}{c_0^2 \Delta}, \quad (13)$$

and the measurement (3) gives

$$\gamma = 2.00 \pm 0.22. \quad (14)$$

It should be observed how narrow the modification is. No field equation is altered; no new field is introduced; no term is added to any Lagrangian. A single coefficient, previously set equal to unity without argument, is restored to the status of a quantity to be measured. We know of no other modification of comparable economy which reproduces the observations.

VI. THE RESTRICTED FORM OF THE PRINCIPLE OF EQUIVALENCE

The principle of equivalence in the form in which it was originally stated [1] asserts that a homogeneous gravitational field is in every respect equivalent to a uniformly accelerated frame of reference. In such a frame a light ray is deflected by exactly the amount (8) with $\gamma = 1$. The value (14) is therefore incompatible with the principle in its unrestricted form.

This is less serious than it appears, for the principle has already once been restricted on compelling grounds. In 1912 [4] it was found that the field equation of the static theory, when corrected so as to satisfy the law of conservation of momentum, no longer admits the homogeneous field as a solution in a finite region, and it was necessary to conclude that “the equivalence principle holds only for infinitely small fields.” That restriction was accepted without a satisfactory reason being known for it. The present restriction is of the same kind and is better motivated, since (11) supplies the reason.

We state the principle henceforth in the following form.

Principle of equivalence

(restricted form). *For ponderable matter, and in infinitely small regions, a gravitational field is equivalent to an acceleration of the frame of reference. For the electromagnetic field the equivalence holds with the coupling γ in place of unity.*

The equality of inertial and gravitational mass, which is the empirical content of the principle and which is verified to a part in 10^9 , refers exclusively to ponderable bodies and is therefore untouched. No experiment has ever compared the gravitational and inertial mass of a quantity of pure radiation, and by (11) no such experiment is possible.

VII. THE GRAVITATIONAL DISPLACEMENT OF SPECTRAL LINES

An atom at rest at potential φ_1 emits, by the restricted principle applied to ponderable matter, with the proper

frequency

$$\nu_1 = \nu_0 \left(1 + \frac{\varphi_1}{c_0^2}\right)^{-1}, \quad (15)$$

the coefficient being unity because the emitting system is ponderable. In a static field the number of wave crests passing any fixed point per unit coordinate time is the same everywhere, whatever the velocity of propagation; the frequency measured at φ_2 is therefore

$$\frac{\Delta\nu}{\nu} = \frac{\varphi_1 - \varphi_2}{c_0^2}, \quad (16)$$

independently of γ . For light emitted at the solar surface and received at the Earth,

$$\frac{\Delta\lambda}{\lambda} = \frac{GM_\odot}{R_\odot c_0^2} = 2.12 \times 10^{-6}, \quad (17)$$

corresponding to 0.64 km s^{-1} in the Doppler measure. The prediction is thus unchanged from that of [1], and the present hypothesis is not exposed to test from this quarter.

VIII. THE CONVENTION FOR THE COMPARISON OF DISTANT CLOCKS

One consequence requires care. If ponderable clocks run with the coefficient unity while light propagates with the coefficient γ , then the velocity of light measured locally, with a ponderable clock and a ponderable rod, is

$$c_{\text{loc}} = c_0 \left(1 + (\gamma - 1) \frac{\varphi}{c_0^2}\right) \neq c_0, \quad (18)$$

in conflict with the requirement that the special theory of relativity hold in the infinitely small.

The conflict is removed by observing that a rod is not measured directly but is *read*, and that it is read by light. Let the reading of a rod illuminated at potential φ be denoted ℓ_{read} and its length ℓ . Then

$$\ell_{\text{read}} = \ell \left(1 + \lambda \frac{\varphi}{c_0^2}\right), \quad (19)$$

and the requirement $c_{\text{loc}} = c_0$ fixes

$$\lambda = \gamma - 1 = 1. \quad (20)$$

With this the local velocity of light is c_0 in every gravitational field, as required, and the special theory of relativity holds locally without restriction.

It is to be emphasized that (19) expresses a property of the process of measurement and not a property of space. The spatial geometry of the static field remains Euclidean; the rods themselves are unchanged; what is corrected is the optical reading of them. Any other view would require an alteration of the line element (4), which,

TABLE II. Observational status after introduction of the optical coupling.

Phenomenon	Observed	Theory	Constant
Light deflection (limb)	$1.75'' \pm 0.19''$	$1.75''$	γ
Radial dependence	$\propto r^{-1}$	$\propto r^{-1}$	—
Tangential component	$0.04'' \pm 0.09''$	0	—
Solar line displacement	$\approx 2 \times 10^{-6}$	2.12×10^{-6}	—
Local velocity of light	c_0	c_0	λ
Equality of masses	1 ± 10^{-9}	1	—
Perihelion of Mercury	$45'' \pm 5''$	$18''$	—

as noted in Sec. V, is excluded by [12]. The situation is analogous to the contraction hypothesis of Lorentz, which likewise concerns the behaviour of measuring apparatus and leaves the underlying space unaffected.

Since λ is fixed by γ and does not enter any observable of first order independently, the theory contains, after the present modification, exactly one new constant.

IX. COMPARISON WITH OBSERVATION

Table II collects the position.

Every entry is now in agreement with observation with the single exception of the perihelion motion of Mercury. The optical coupling does not act upon a ponderable planet, and the value $18''$ per century obtained from the field equations of [5] is therefore unaffected by anything in the present paper. The residual $27''$ must be sought elsewhere. A treatment will be given in a following communication [13].

X. CONCLUDING REMARKS

The measurements reported here refute a numerical prediction which has stood since 1911 and which was regarded, by the present author among others, as one of the securest consequences of the theory of the static field. It is worth recording that the refutation cost the theory very little. The field equations stand; the line element stands; the law of motion of ponderable bodies stands; the displacement of spectral lines stands; the equality of inertial and gravitational mass stands. What has fallen is an assumption which was never stated, namely that the electromagnetic field and ponderable matter respond to gravitation with the same coefficient, and which the vanishing of the electromagnetic trace (11) shows to have been implausible from the beginning.

That a theory should be capable of absorbing so severe a contradiction with so small an alteration is, in our judgement, evidence in its favour and not against it. Nature shows us only the tail of the lion. But I have no doubt that the lion belongs to it, even though he cannot at once reveal himself because of his enormous size.

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Appendix A: Plate constants and systematic errors

The systematic uncertainty in (3) arises from four sources.

(i) *Change of plate scale between epochs.* The eclipse plates were exposed at an ambient temperature of 27°C , the comparison plates at -3°C . The resulting change of focal length alters the scale by 2.1×10^{-5} , which is absorbed by the constants b and f of (2) but is degenerate with α_0 to the extent that the star field is not symmetric about the solar centre. Estimating this degeneracy from the actual distribution of the 50 images gives $\pm 0.09''$.

(ii) *Differential refraction.* The altitude of the Sun at totality was 53° , that of the comparison field 61° . The differential refraction across the 2° field is $0.11''$ and is radial to 4 per cent. The residual contributes $\pm 0.05''$.

(iii) *Emulsion shift.* Plates from the same batch, developed together, show relative displacements of up to $0.06''$ over the plate. Averaging over seven plates reduces this

to $\pm 0.03''$.

(iv) *Coma.* The images near the edge of the field are elongated radially by $0.3''$. Since the elongation is radial it cannot be separated from the effect sought by symmetry arguments alone, and was determined instead from the comparison plates, where the same elongation appears with the same dependence on field position and no deflection is present. The correction so obtained is $-0.08''$ at the limb with uncertainty $\pm 0.09''$.

The four contributions combined in quadrature give $\pm 0.14''$.

Appendix B: On the numerical value of γ

The measurements give $\gamma = 2.00 \pm 0.22$, and it is natural to ask whether the value 2 is exact and whether a reason can be assigned for it.

We know of no principle from which the value follows. The theory as it stands admits any value of γ ; the number must be taken from experiment, as the ratio of the electron's charge to its mass is taken from experiment.

We record, without attaching weight to it, the following circumstance. The electromagnetic field in free space possesses two independent states of polarization, whereas a ponderable particle in the static theory couples to the field through a single scalar. Should the coupling be additive over the independent states, the factor 2 would follow. We have not succeeded in giving this remark a precise form, and it is possible that the coincidence is accidental. A determination of γ to one part in a hundred, which a further eclipse should provide, would decide whether the value is exactly 2 or merely near it.

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